

ANNEXURE A – CASE STUDY

1. Satellite Image Registration Using Harris Corner Detection

Case Study

A remote sensing system is used to align satellite images of the same geographical area captured at different times. Initially, the system uses edge detection to find matching regions. However, changes in illumination, orientation, vegetation, shadows, and background details make edge-based matching unreliable. Therefore, the system replaces edge detection with Harris Corner Detection.

Limitations of Edge Detection

- Illumination sensitivity – Changes in sunlight and weather can change edge intensity.
- Background details – Vegetation, clouds, shadows, and other background elements can create unwanted edges.
- Poor localization – Edges are continuous lines rather than distinctive points.
- Difficult matching – Corresponding edges may have different shapes or intensities in images captured at different times.
- Orientation changes – Changes in camera or satellite orientation can make edge correspondence difficult.
- Noise sensitivity – Small variations can produce additional false edges.

How Harris Corner Detection Improves the System

Harris Corner Detection identifies points where image intensity changes significantly in two directions simultaneously. These points are more distinctive than ordinary edges and can be used as control points for image registration.

- Building corners
- Road intersections
- Bridge corners
- Distinctive structures
- Sharp geographical features

Comparison

Parameter	Edge Detection	Harris Corner Detection
Output	Edges/lines	Distinctive points
Localization	Moderate	Better
Matching	Difficult	Easier
Illumination changes	Sensitive	More stable
Orientation changes	Less reliable	Better for local point matching
Image registration	Limited	More suitable
Feature distinctiveness	Low–Moderate	High at corners

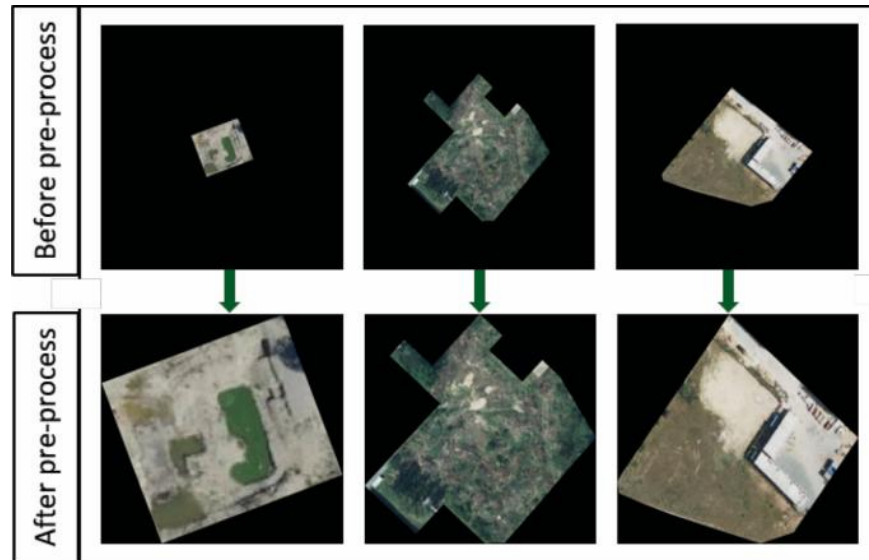
Analysis

Harris Corner Detection provides localized feature points that are easier to identify and compare between two satellite images. Instead of trying to match entire edges, the system can match corresponding corners and use them as reference points for estimating the transformation between images.

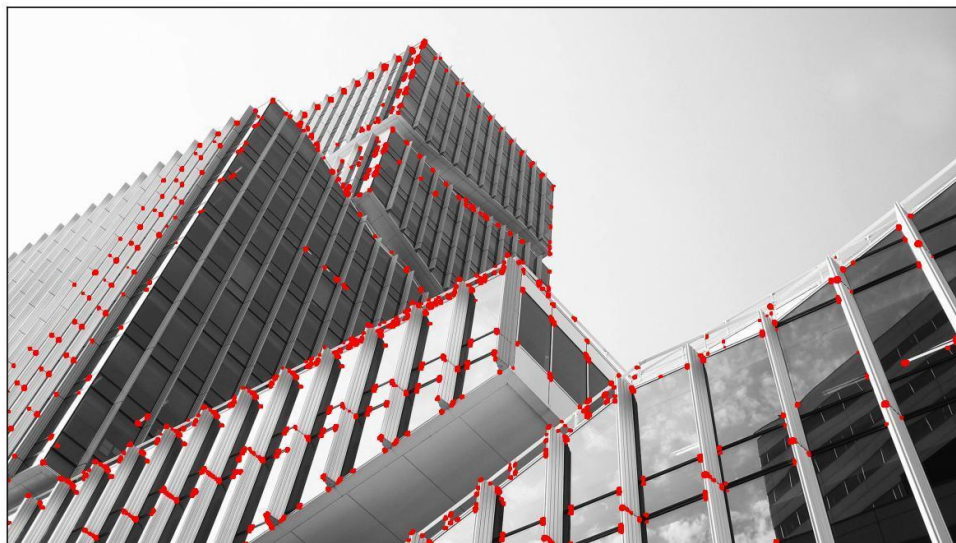
For example, if a road intersection appears in both satellite images, Harris can identify the intersection as a distinctive corner structure. Even if the image has some illumination or background variation, the location of the structural feature can remain useful for registration.

Images to Paste

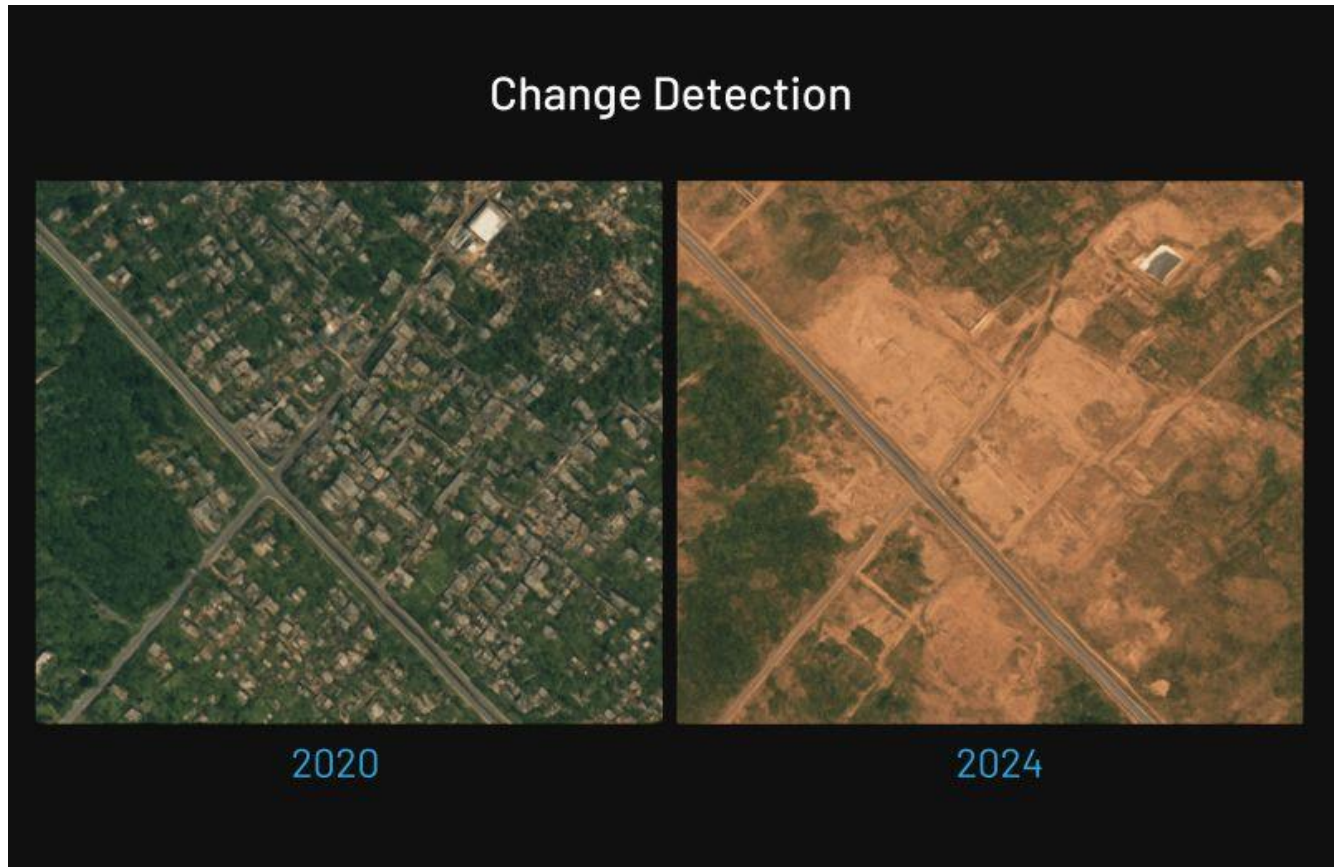
- Fig. 1: Satellite Images Captured at Different Times – paste two input satellite images here.



- Fig. 2: Harris Corner Features for Image Registratio – paste the Harris corner detection output here.



- Fig. 3: Registered Satellite Images – paste the final aligned/registered result here.



Conclusion

Edge detection alone is not reliable for satellite image registration because edges are sensitive to illumination, noise, background changes, and orientation. Harris Corner Detection provides distinctive and localized feature points that can be matched between images. Therefore, Harris Corner Detection is a more appropriate feature-localization technique for reliable image registration under varying imaging conditions.

2. Image Representation for Real-Time Surveillance

Case Study

An autonomous surveillance system continuously captures high-resolution color images to detect vehicles and pedestrians in real time. Processing full-color images requires more data and computational resources. The engineers therefore consider converting the images into grayscale or binary representations before feature detection.

Color Representation

A color image generally contains three channels: Red, Green, and Blue. Therefore, a high-resolution RGB image contains considerably more data than a single-channel grayscale image.

Advantages

- Preserves complete color information.
- Useful when object color is important.
- Can distinguish objects with similar shapes but different colors.

Disadvantages

- Higher memory requirement.
- More computational operations.
- Slower processing.
- More data for feature extraction.

Grayscale Representation

A grayscale image contains a single intensity channel. A common conversion is approximately: $\text{Gray} = 0.299R + 0.587G + 0.114B$.

Advantages

- Reduces the image from three channels to one.
- Requires less memory.
- Reduces computational complexity.
- Provides sufficient intensity information for many edge detectors.
- Works well with feature detectors such as Harris and SIFT.
- Faster for real-time processing.

Disadvantages

- Color information is lost, so objects distinguished mainly by color may become harder to separate.

Binary Representation

A binary image contains only two values, usually 0 (black) and 255 (white). Binary images can be created using thresholding.

Advantages

- Very low computational complexity.
- Very small representation.
- Useful for simple object segmentation.
- Fast processing.

Disadvantages

- Significant information loss.
- Fine image details may disappear.
- Sensitive to lighting and threshold selection.
- Small features may be lost.
- Not suitable for all feature detection algorithms.

Comparison

Feature	Color	Grayscale	Binary
Channels	3	1	1
Data size	High	Low	Very low
Processing speed	Slow	Fast	Very fast
Color information	Yes	No	No
Edge detection	Good	Very good	Good
Feature extraction	Good	Very good	Limited
Detail preservation	High	Good	Low
Real-time suitability	Moderate	High	High for simple scenes
Lighting robustness	Depends	Better for many intensity-based tasks	Low–Moderate

Analysis

For vehicle and pedestrian detection, grayscale representation is the best general-purpose choice when the feature detector mainly depends on intensity, edges, corners, or local texture.

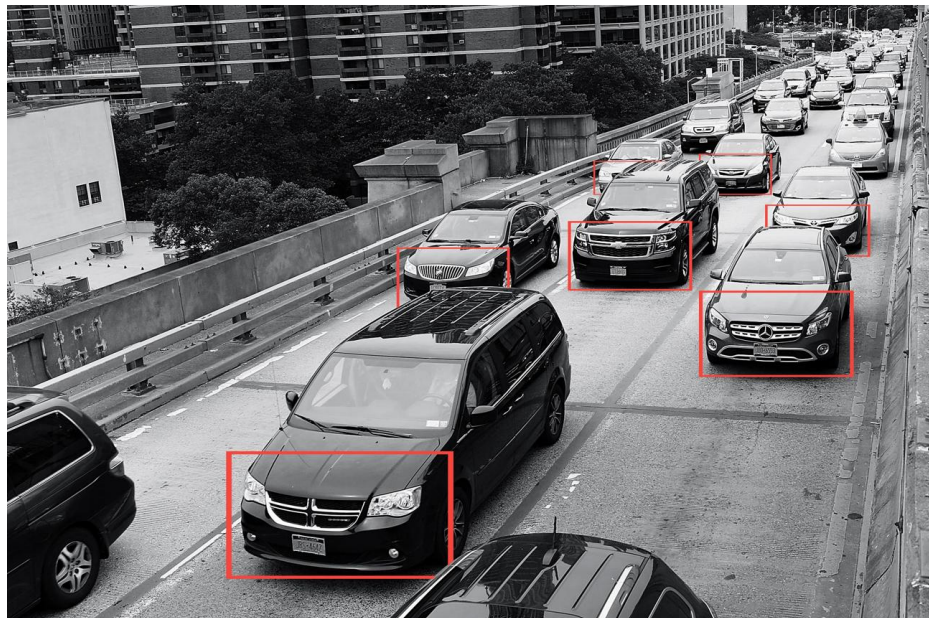
Color images contain additional information, but processing three channels increases computational requirements. Binary images are faster still, but converting an image to binary removes many details that may be useful for reliable feature extraction.

Recommended Representation

For the given real-time surveillance application, grayscale is the most appropriate representation because it provides a good balance between computational efficiency, edge detection, feature extraction, object detail, and processing speed. Binary representation should be used when the scene is simple and reliable segmentation is possible.

Images to Paste

- Fig. 4: Original Color Image – paste the original surveillance image here.



-
- Fig. 5: Grayscale Representation – paste the grayscale version here.



- Fig. 6: Binary Representation – paste the thresholded binary version here.



(a) Raw image

(b) Detected objects

- Fig. 7: Edge Detection on Different Image Representations – paste the comparison output here.



Conclusion

Image representation has a major effect on computer vision performance. Color images preserve the most information but require greater computational resources. Binary images provide fast processing but lose significant information. Grayscale images provide the best balance between computational efficiency and useful visual information. Therefore, grayscale is recommended for the surveillance .